

FUTURE-READY (Clockwise from left) The experimental Hyperloop tube; the pod; Union minister Ashwini Vaishnaw inspects the pod model with a student team at IIT Madras in Chennai

THE HYPERLOOP PROMISE

The concept is an ultra-high-speed public transportation system where passengers travel in autonomous electric pods through near-vacuum tubes. Often called the fifth mode of transport, it reduces air resistance to allow capsules to reach speeds exceeding 1,000 km/h



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to 50 km, a scale no other country has attempted. It's a bold leap: to set global benchmarks in propulsion, levitation, safety and cost efficiency.

So, what exactly is the Hyperloop? Imagine stepping into a pod roughly the size of a hatchback car, which accelerates inside a sealed vacuum tube to speeds approaching 1,200 km per hour, far faster than a commercial jet. That's the promise. But how is it done? With the air removed, the vacuum tube eliminates drag. Electromagnetic levitation lifts the pod off the surface, and a linear electric motor provides thrust. The result? A near-frictionless glide at breathtaking velocity. "We are constructing a sub-scale passenger cabin, a concrete tube rather than a steel tube, which is going to ensure over 80 per cent cost savings," says Prem Mukkannavar, a third-year civil engineering student. "We are also working on a pioneering end-to-end booster-cruiser technology."

Hyperloop became the big buzzword post-2013, following the publication of 'Hyperloop Alpha', a white paper by tech billionaire Elon Musk. However, the concept has more than a century of history. It runs the gamut from George Medhurst's 18th-century idea of air-powered tubes, to pneumatic rail networks in 19th century London. That early, experimental phase faded away. But Musk breathed new life into the concept, giving it some tech optimism.

THE INDIAN INVENTION

Avishkar Hyperloop was launched in 2017. Its current testbed, likely the world's largest student-led Hyperloop initiative, can replicate the vacuum conditions needed to remove drag and enable a pod to zip using electromagnetic propulsion inside a two-metre-diameter tube. Using computational topology optimisation, a process that allows computers to identify the most efficient structural shape by eliminating non-essential material, the team reduced the chassis mass by 46 per cent with-